Sparse-GC Recovery and Fuzz-Regression Checkpoint

Tactix Development Project

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Executive summary

The open P4-CVG `mod4block` investigation reached a strong but held candidate. The conflict-1010 trajectory split was caused by sparse Reduce collection resetting the replacement-search cursor of clauses in the preserved irredundant prefix. Kissat retains that cursor and the existing prefix watch order. The candidate makes Tactix preserve both, including when earlier garbage changes prefix arena offsets.

The result is material: the trace-off production pin solves `mod4block` with an independently verified SAT model in 31.87 seconds, where the preceding shipped pin timed out at 120 seconds; Kissat takes 18.60 seconds. A matched trace reproduces the target cursor sequence and keeps decisions and learned sets aligned through conflict 1012.

The candidate is not accepted. The required quick fuzz run fails tracked corpus cases 004 and 005 before reaching the 300 generated inputs. One failure is an ASan out-of-bounds read in Sweep after Factor creates extension variables; the other passes an out-of-range literal to proof serialization. These are consistent with one invalid arena-reference lifecycle, but that attribution is not yet proved. In addition, the earlier `tr-n300-s055` fixture solves proof-clean in 36.43 seconds at a 120-second limit but misses its required 30-second cumulative guard.

1 Implementation decision

Sparse collection now treats `watch_start` as a semantic boundary rather than only an optimization hint:

- copied prefix clauses retain `Clause::searched`;
- live prefix watch entries retain order and blocker while their Refs are forwarded around removed prefix garbage;
- the relocated tail boundary is computed after compaction; and
- only tail clauses reset their cursor and receive rebuilt watches.

Optional paired trace fields `TACTIX_EVENT_TRACE_WATCH_POSE` and `KISSAT_EVENT_TRACE_WATCH_POSE` expose the physical literal pose and cursor. They are compiled only into trace builds. Focused tests cover cursor preservation, prefix garbage, reference forwarding, tail rewatching, and trace output.

2 Evidence

Check	Result
Matched trace	Job `1607099`; both models verified. Target scans match `7->8->9->2`; decisions and learned sets align through conflict 1012.
Production	Job `1607102`; Tactix 31.874s and Kissat 18.603s, both SAT/model verified. Prior pin timed out at 120s.
Release suite	Immutable SHA-256 `1e8e5ef213533fd0d1ca8e1e8a245d0cbb865d04ec2502afa5a11a7ef3` 69/69 CTests passed, including proof/model and trace-zero-overhead checks.
P4A guards	Jobs `1607105`, `1607104`, and `1607217`: n260 6.074s proof-clean; n280 14.348s proof-clean; n300 36.425s proof-clean but over the 30s guard.
n300 bisection	Job `1607222` at `95bcea4`: 28.841s. Jobs `1607223`/`1607227` at `c5a98e4`: timeout at 30s, 35.754s at 120s. The regression predates this candidate.
Fuzz gate	Job `1607229` fails corpus 004/005. Jobs `1607232` and `1607233` isolate the Sweep ASan read and proof-map assertion.

The detailed evidence record is `benchmarks/reports/2026-08-10-mod4block-sparse-gc-checkpoint.md`.

3 CI quota change

The GitHub Actions workflow no longer runs the full sanitizer, checker, and fuzz matrix on every push and pull request. It is scheduled weekly (Monday at 06:17 UTC), remains manually dispatchable, and cancels an older run for the same ref. This deliberately bounds routine Actions-minute consumption while retaining an explicit milestone gate.

4 Risks and exact next actions

1. Reproduce corpus 004 on Slurm with per-rule tracing and identify the first invalid arena reference. The attempted trace submission at wrap was blocked by the workspace sandbox from opening the Slurm controller socket; no job remains active.

2. Add a focused Factor/Reduce/Sweep interleaving regression, repair the owning reference lifecycle, and rerun corpus cases 004/005 followed by the exact 300-case quick fuzz command.

3. Return immediately to P4-CVG fixture order. Continue the already localized `c5a98e4` n300 trajectory regression, then rerun all three P4A guards and the `mod4block` production/proof/model gate.

4. Only after `mod4block` closes proceed to `mp1`; do not skip the cumulative fixture guards.

The Slurm socket denial is an execution-sandbox restriction, not evidence of a repository, cluster, or solver security compromise.